

Common-Coordinate Transport of Riesz Clouds

Summable Resolvent Defects and Fixed-Rank Limits

Prime Dynamics Theory Program

July 2026

Abstract

RH-162–168 give fixed-scale packet-to-Riesz certificates. An all-level interface also needs a coherent comparison of Riesz clouds living at different mesh and noise scales. After embedding two consecutive operators in a common Hilbert space, we prove

$$\|\Pi_{j+1} - \Pi_j\| \leq \frac{|\Gamma|}{2\pi} M_{j+1} M_j \|A_{j+1} - A_j\|,$$

where one contour Γ lies in both resolvent sets and M_j are its resolvent bounds. A step bound below one yields an invertible range transport with explicit inverse norm. If the step bounds are summable, the Riesz projections converge in operator norm to an idempotent of the same finite rank. A 512-pair Hermitian audit has no inequality failure. The theorem is restricted to fixed rank; a moving determinant cloud has growing rank and requires the shellwise correction developed next.

1 The common-coordinate input

Operators at different Ulam dimensions cannot be subtracted until maps into one ambient space have been chosen. Let $\mathcal{H}_j$ be the scale spaces, let $J_j : \mathcal{H}_j \rightarrow \mathcal{K}$ be isometries, and suppose extensions $\tilde{A}_j \in \mathcal{B}(\mathcal{K})$ have been specified. Their action outside $J_j \mathcal{H}_j$ is part of the construction; arbitrary zero extension can introduce spurious spectrum on a contour.

For the theorem, write A_j for these common-space operators. Let one rectifiable Jordan curve Γ satisfy

$$\Gamma \subset \rho(A_j) \cap \rho(A_{j+1}), \quad \sup_{\Gamma} \|(z - A_j)^{-1}\| \leq M_j.$$

Define

$$\Pi_j = \frac{1}{2\pi i} \int_{\Gamma} (z - A_j)^{-1} dz.$$

2 Consecutive transport theorem

Theorem 1 (Common-contour Riesz transport). *If $\|A_{j+1} - A_j\| \leq \eta_j$, then*

$$\|\Pi_{j+1} - \Pi_j\| \leq \beta_j := \frac{|\Gamma|}{2\pi} M_{j+1} M_j \eta_j.$$

Proof. The resolvent identity gives

$$(z - A_{j+1})^{-1} - (z - A_j)^{-1} = (z - A_{j+1})^{-1}(A_{j+1} - A_j)(z - A_j)^{-1}.$$

Integrate around Γ and take norms [Kato, 1995]. $\square$

Corollary 2 (Stable range map). *Suppose Π_j and Π_{j+1} have equal finite rank and $\beta_j < 1$. Then*

$$\Pi_{j+1}|_{\text{ran } \Pi_j} : \text{ran } \Pi_j \longrightarrow \text{ran } \Pi_{j+1}$$

is invertible, and its inverse norm is at most $(1 - \beta_j)^{-1}$.

Proof. For $x \in \text{ran } \Pi_j$,

$$\|\Pi_{j+1}x\| \geq \|x\| - \|(\Pi_{j+1} - \Pi_j)x\| \geq (1 - \beta_j)\|x\|.$$

Thus the map is injective; equal finite dimensions give surjectivity and the inverse bound. $\square$

3 Summable fixed-rank limit

Theorem 3 (Norm-convergent Riesz chain). *Assume the same finite rank r is enclosed at every scale and*

$$\sum_{j=1}^{\infty} \beta_j < \infty.$$

Then Π_j converges in operator norm to a bounded idempotent Π_{∞} of rank r . The tail obeys

$$\|\Pi_{\infty} - \Pi_n\| \leq \sum_{j=n}^{\infty} \beta_j.$$

Proof. The telescoping inequality makes (Π_j) norm-Cauchy. Completeness of $\mathcal{B}(\mathcal{K})$ gives a limit, and multiplication is norm continuous, so $\Pi_{\infty}^2 = \Pi_{\infty}$. For sufficiently large j , $\|\Pi_j - \Pi_{\infty}\| < 1$; the close-projection restriction argument gives equal rank. The tail estimate follows from telescoping. $\square$

Summability is stronger than $\eta_j \rightarrow 0$. Resolvent conditioning can grow fast enough that $M_j M_{j+1} \eta_j$ is not summable even when operators look close entrywise. This is the all-level analogue of the fixed-scale pseudospectral wall.

4 Moving contours and homotopy compatibility

Consecutive physical shells may use different contours. They can be compared by Theorem 1 only after a common representative is chosen.

Proposition 4 (Contour deformation ledger). *Let Γ_0 and Γ_1 be homologous closed contours in one connected open subset of $\rho(A)$, enclosing the same spectral component. Then their Riesz integrals are equal. Consequently, if both consecutive operators are resolvent-free throughout a common deformation corridor, either boundary may be used in the transport estimate.*

Proof. The operator-valued resolvent is holomorphic on the corridor. Cauchy's theorem makes the two contour integrals equal [Kato, 1995]. $\square$

This adds a concrete schedule obligation. Merely labeling two independently chosen clouds by the same shell index does not prove they enclose the same spectral component; a joint resolvent corridor or an equivalent coefficient identification is required before telescoping.

5 Vanishing steps are not enough

The summability hypothesis in Theorem 3 cannot be replaced by $\beta_j \rightarrow 0$ in this proof architecture.

Proposition 5 (Slow rotating packet witness). *There is a sequence of rank-one orthogonal projections on $\mathbb{C}^2$ such that*

$$\|P_{j+1} - P_j\| \rightarrow 0$$

but (P_j) does not converge.

Proof. Let

$$\theta_j = \sum_{k=1}^j \frac{1}{k}, \quad \text{quada}_j = (\cos \theta_j, \sin \theta_j), \quad P_j = u_j u_j^*.$$

Then $\|P_{j+1} - P_j\| = |\sin(1/(j+1))| \rightarrow 0$. The angle θ_j keeps traversing the projective circle because its increments tend to zero while their sum diverges. Hence the ranges visit separated angular sectors infinitely often and do not converge. $\square$

Taking $A_j = P_j$, a fixed circle around the eigenvalue one gives uniformly bounded resolvents and Riesz projection P_j . Thus even an exact uniformly gapped spectral model can drift forever when the common-coordinate defects are nonsummable.

6 Conditioning of the composed range transport

Let

$$L_j = \Pi_{j+1}|_{\text{ran } \Pi_j}$$

be the step maps from Corollary 2. Their composition transports one initial cloud frame through the chain without making a new gauge choice at every scale.

Proposition 6 (Finite cumulative inverse condition). *If $\beta_j \leq \beta_* < 1$ and $\sum_j \beta_j < \infty$, then*

$$\prod_j \|L_j^{-1}\| \leq \prod_j (1 - \beta_j)^{-1} < \infty.$$

More explicitly,

$$\log \prod_{j=n}^N (1 - \beta_j)^{-1} \leq \frac{1}{1 - \beta_*} \sum_{j=n}^N \beta_j.$$

Proof. Corollary 2 gives the first finite-product inequality. For $0 \leq x \leq \beta_*$, $-\log(1 - x) \leq x/(1 - \beta_*)$. Sum this scalar inequality and exponentiate. $\square$

Thus the same summability that yields a projector limit also prevents the canonical stepwise coordinates from acquiring infinite inverse condition. This is stronger information than pairwise rank agreement and is relevant when shell coefficients or marked vectors must be transported coherently.

7 Audit and boundary

The finite audit rotates a three-dimensional spectral packet inside eight- dimensional Hermitian matrices while keeping a unit-circle spectral gap. It evaluates the exact normal resolvent bound, the operator defect, and the projector distance for 512 pairs. No distance exceeds Theorem 1's envelope.

RH-169 proves the correct fixed-rank common-coordinate theorem. It does not construct the physical embeddings or extensions, prove summability, or handle the increasing degree of the RH-80 cloud. Therefore physical R, Gate A, and all later Hilbert–Polya interfaces remain open.

References

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.